

## **CHANGING THE SETUP OF WORK SPINDLE**

Workhead with shift function  
Dead spindle workhead with NC shift  
Workhead with longitudinal positioning

Read the operation manual and become fully familiar with its contents before operating this machine. Store this operation manual at a location set aside for it so that it can be accessed immediately whenever necessary.

## Handing Over the Machine

Strictly observe the following points when the machine is handed over.

- Both this manual and the other accompanying documents must be handed over to the new owner.
- Refer to the chapter on “Warning labels” and check all the warning labels.
- If any warning label or manual is missing, contact JTEKT Corporation.

Contact the JTEKT Corporation when the machine is handed over.

Note the following contents.

This machine may be classified as “strategic goods” under the Foreign Exchange and Foreign Trade Control Act.

In order to export this machine from Japan, permission must be obtained in compliance with this Act.

This machine has been manufactured to be in compliance with laws, standards, and regulations, and standards of the country or region where it is used.

Accordingly, it cannot be exported to a country or region with different laws, standards, and regulations.

## Guarantee

### Cost-free Repairs

- Faults and failures that occur while the machine is being used under normal conditions of use in compliance with the handling criteria (listed in the operation manual), and the operation manuals, affixed labels, etc., are repaired free of cost within the term of the guarantee.
- The coverage of the guarantee is limited to repair to the delivered machine body only. Other damage caused directly or indirectly is not covered by the guarantee.
- The term of the guarantee is one year, commencing at delivery and acceptance.

### Chargeable Repairs

Repairs done after the term of the guarantee, and those done within the term of the guarantee under the following conditions, are chargeable.

- Failure and damage due to misuse by the client, or due to modifications or repairs undertaken without the approval of JTEKT Corporation.
- Failure and damage caused by unpredictable events such as fire, acts of God, lightning, abnormal voltages, and so on.
- Failure and damage arising from toppling of the machine during transportation or shifting of the machine by the client, or due to inappropriate handling.
- Failure and damage caused by not complying with the handling criteria.
- Wear and tear, abrasion, and deterioration of consumable parts in normal use.
- Failure or damage caused by oil being not designated even though oil to be used is designated. Failure and damage caused by oil, the viscosity of which is not proper, or which has been used away from its intended use purpose.

## About the Manual

- This operation manual is subject to change without notice due to our policy of continually improving the machine and the operation manual.  
This can lead to some discrepancies between the manual and the machine: please bear this in mind.  
Where there is a difference between the machine and the text of the operation manual, or if the meaning of any part of the text is unclear, please contact JTEKT Corporation.
- Reproduction of this manual is prohibited without prior permission.

## HOW TO READ THIS MANUAL

### (a) Representation of Safety Caution Marks

This manual contains the following indications to enable you to use the machine safely, and in order to prevent hazards to personnel and damage to property.



Failure to observe items in this category will result in death or severe injury.



Failure to observe items in this category can lead to death or severe injury.



Failure to observe items in this category can lead to injuries or property damage.

In addition to the above, the following indications are also used in this manual.



This is an item that requires attention during machine operation or other work. Failure to observe such items can lead to damage to the machine.



This is information for reference during machine operation and other work.



This is information for reference during machine operation and other work.

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## HOW TO READ THIS MANUAL

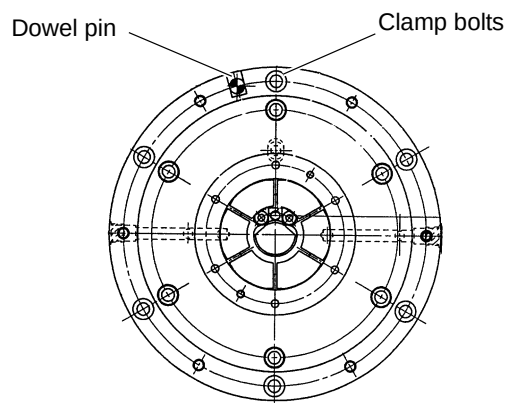
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# 1. CHANGING THE SETUP OF WORK SPINDLE

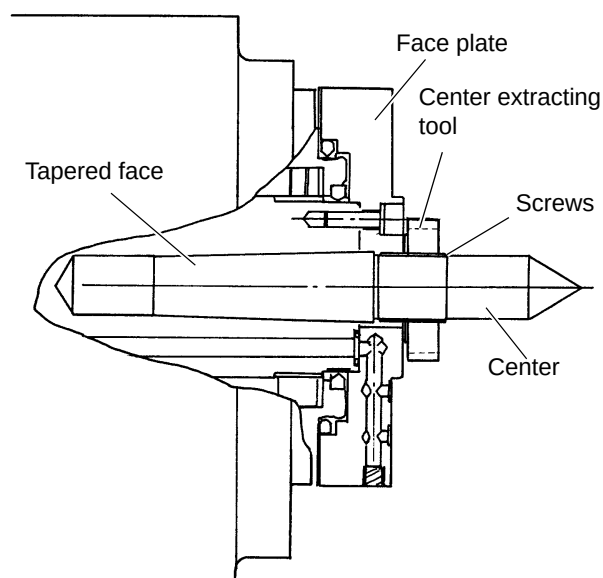
Explains the specifications mounting the chuck to workhead as an example.

## 1.1 Changing the Work Spindle Center

- (1) Place the [OPERATION MODE] selection switch in the [MANU.] position.
- (2) Press the [GR.WHEEL SPINDLE STOP] button and stop the wheel rotation.
- (3) Open the front door.
- (4) Press the [EMERGENCY STOP] button.
- (5) Pull out the dowel pin.
- (6) Loosen the chuck clamp bolts (M6, 6 pcs.) by turning them counterclockwise.  
Remove the bolts and detach the chuck from the work spindle.



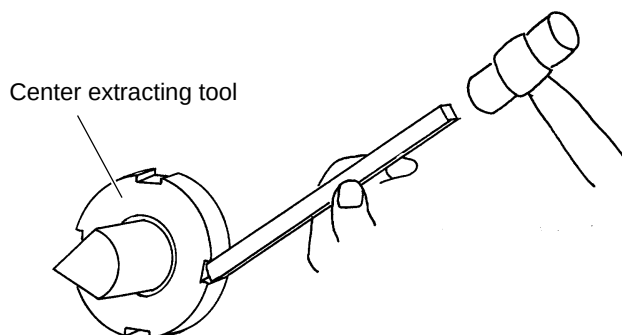
- (7) Fit the center extraction tool onto the threads of the center by turning it clockwise.  
After the center extraction tool contacts the end face of the tailstock spindle, turn it further clockwise; the center will come off from the tailstock spindle.



### <NOTE>

Since the center is fitted into the center hole by means of a tapered face, set a bar on the center extraction tool and tap it lightly with a hammer as shown in the figure.

- (8) Clean the center hole.
- (9) Wind thin paper on the new center and insert it into the center of the tailstock spindle.  
Turn the center several times to clean the inside face of the center hole.
- (10) Remove the paper from the center and insert the center into the center hole.



# 1. CHANGING THE SETUP OF WORK SPINDLE

## <NOTE>

When inserting the center, face the recessed section at the end of the center to the direction indicated by the arrow symbol (↓).

The chuck cannot be mounted if the center is inserted in the opposite direction.

(11) Mount the chuck. Tighten the chuck clamp bolts (M6, 6 pcs.) by turning them in the clockwise direction.

(12) Insert the dowel pin.

(13) Tighten the chuck clamp bolts.

(14) Mount the lever type dial gauge.

Fix the stand to the table or workhead and touch the center tip with the dial gauge.

(15) Unlock the [EMERGENCY STOP] button by turning it clockwise.

(16) Press the [MASTER ON] button.

(17) Place the [CNC MODE] selection switch in the [HANDLE/JOG] position.

(18) Select [C] axis by the [Axis select] switch on the manual pulse generator.

(19) Rotate the C-axis (work spindle) by the pulse handle to measure run-out of the front end of the center.

Check that the runout at the center end is within 2 $\mu$ m TIR.

## <NOTE>

It seems that there is foreign matter or scratch in the center hole if the runout at the center end exceeds 2 $\mu$ m TIR. Detach again the chuck and center, and clean the center hole.

It seems that the bearing of work spindle is worn if there is runout in the center hole.

(20) Set the dial indicator at the clamp face of the chuck.

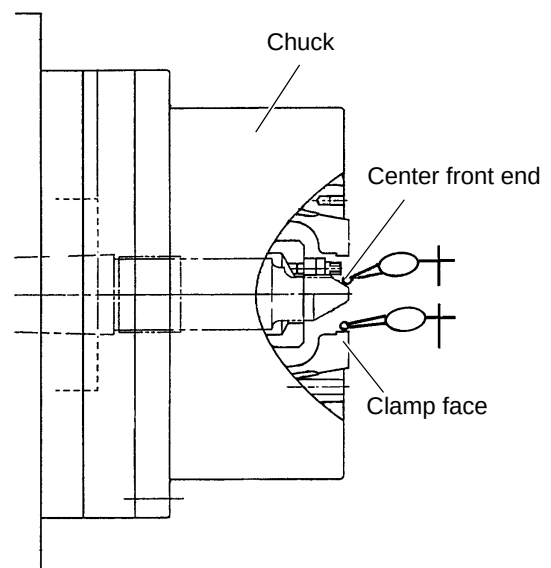
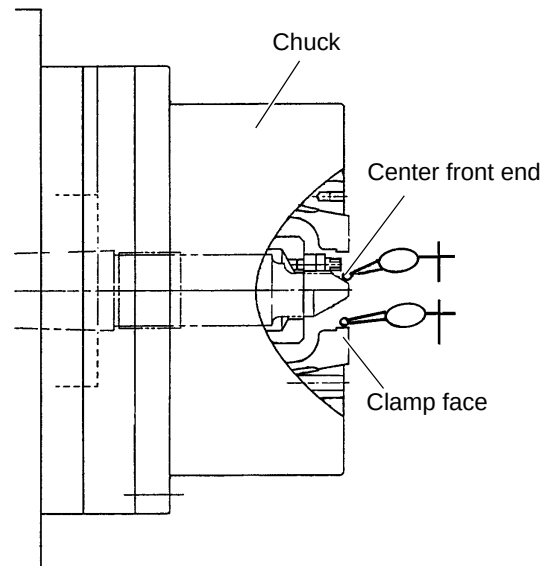
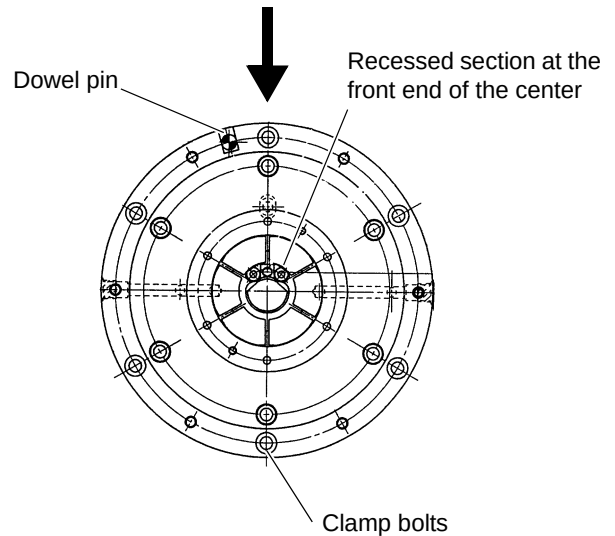
(21) Rotate the C-axis (work spindle) by the pulse handle to measure run-out of the chuck clamp face.

Make sure that the run-out is within 0.01mm TIR.

## <NOTE>

If the measured amount of run-out is larger than 0.01 mm TIR, the collet chuck mounting face may be scored or there may be foreign matter on the collet chuck mounting face. In this case, remove the collet chuck and clean the collet chuck mounting face again.

If the collet chuck clamp face has been worn, change the collet chuck.



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### 1.2 Adjusting the Work Spindle Position

- (1) Place the [OPERATION MODE] selection switch in the [MANU.] position.
- (2) Press the [GR.WHEEL SPINDLE STOP] button and stop the wheel rotation.
- (3) Open the front door.
- (4) Prepare the tooling diagram supplied with the machine.  
Check the distance between the workhead end face and the chuck end face specified in the tooling diagram.
- (5) Unlock the [EMERGENCY STOP] button by turning it clockwise.
- (6) Press the [MASTER ON] button.
- (7) Place the [CNC MODE] selection switch in the [HANDLE/JOG] position.
- (8) Select [WL] axis by the [Axis select] switch on the manual pulse generator.
- (9) Place the [Multiplication Ratio] selection switch in the [x10] or [x100] position.
- (10) Move the work spindle (WL axis) by operating the pulse handle to the position checked in the tooling diagram.
- (11) Click the [CURRENT POSITION] button at the screen. The CURRENT POSITION DISPLAY window opens.  
Find the WL axis position and record it.
- (12) Place the [CNC MODE] selection switch in the [MDI] position.
- (13) Place the [PROGRAM EDIT] selection switch in the [ON] position.
- (14) Click the [CLOSE] button of the screen at the current position display window. The current position display window is closed.
- (15) Click the [CHGOVR MENU] button at the screen. The CHGOVR MENU screen is displayed.
- (16) Click the [GRINDING DATA] button at the screen. The GRINDING DATA screen is displayed.
- (17) Click the [EDIT] button at the screen. The GRINDING DATA EDIT screen is displayed.
- (18) Click the WORK NUMBER field of the workpiece for which setup is changed. The WORK NUMBER field is highlighted.
- (19) Click the [SELECT] button at the screen. The GRINDING DATA EDIT screen is displayed.
- (20) Click the data field of B054 (WORK SPNDL ADV POS). The key-panel window opens. Enter the current position registered at the step (11) at the KEY PANEL window.
- (21) Click the [NEXT ITEM] button of the screen. The next item is displayed.
- (22) Click the [REGISTER] button of the screen. The grinding data is registered and the grinding data edit screen is exited.
- (23) Place the [PROGRAM EDIT] switch in the [OFF] position.

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## 1.3 Changing the Setup for the Driving Dog

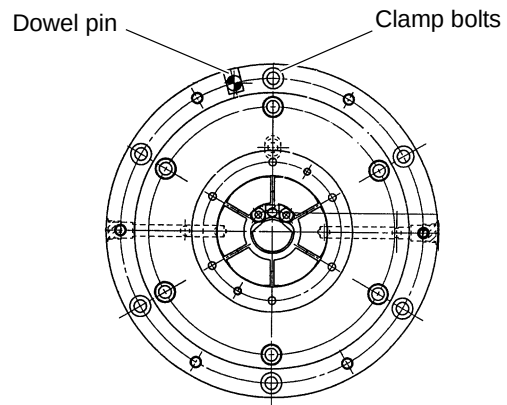
### (a) Changing and Adjusting the Collet Chuck

- (1) Place the [OPERATION MODE] selection switch in the [MANU.] position.
- (2) Press the [GR.WHEEL SPINDLE STOP] button and stop the wheel rotation.
- (3) Open the front door.
- (4) Press the [EMERGENCY STOP] button.
- (5) Remove the dowel pin.
- (6) Turning counterclockwise the clamp bolts ((M6, 6 pcs.) of collet chuck, loosen and remove the bolts.

Remove the collet chuck from the face plate.

#### <NOTE>

An O ring is set in the oil hole at the faceplate.  
Pay attention so that you will not lose the O ring when remove the chuck.

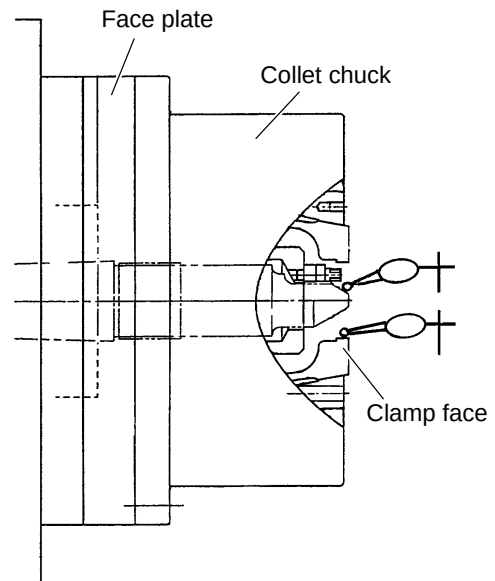


- (7) Clean the mounting faces of collet chuck and face plate, and apply an oil stone lightly.
- (8) Mount the collet chuck to the face plate. Turn clockwise the clamp bolts (M6, 6 pcs.) of collet chuck and tighten them lightly.
- (9) Insert the dowel pin.
- (10) Tighten the clamp bolts of chuck firmly.
- (11) Mount the lever type dial gauge.  
Fix the stand to the table or workhead and touch the center tip with the dial gauge.
- (12) Unlock the [EMERGENCY STOP] button by turning it clockwise.
- (13) Press the [MASTER ON] button.
- (14) Place the [CNC MODE] selection switch in the [HANDLE/JOG] position.
- (15) Select [C] axis by the [Axis select] switch on the manual pulse generator.
- (16) Rotate the C-axis (work spindle) by the pulse handle to measure run-out of the chuck clamp face.

Make sure that the run-out is within 0.01mm TIR.

#### <NOTE>

If the measured amount of run-out is larger than 0.01 mm TIR, the collet chuck mounting face may be scored or there may be foreign matter on the collet chuck mounting face. In this case, remove the collet chuck and clean the collet chuck mounting face again.  
If the collet chuck clamp face has been worn, change the collet chuck.

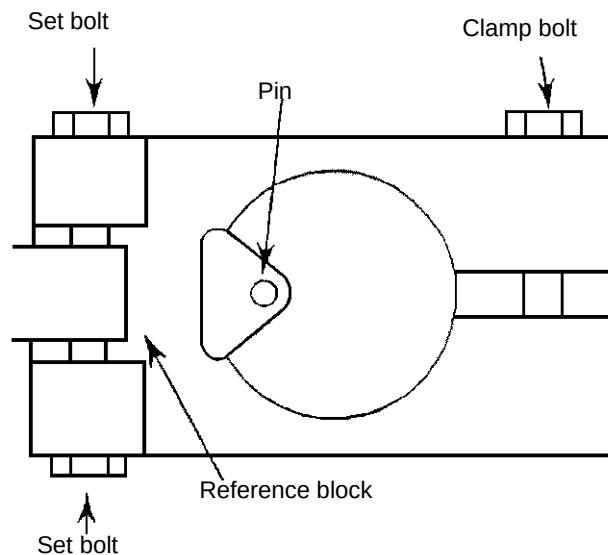


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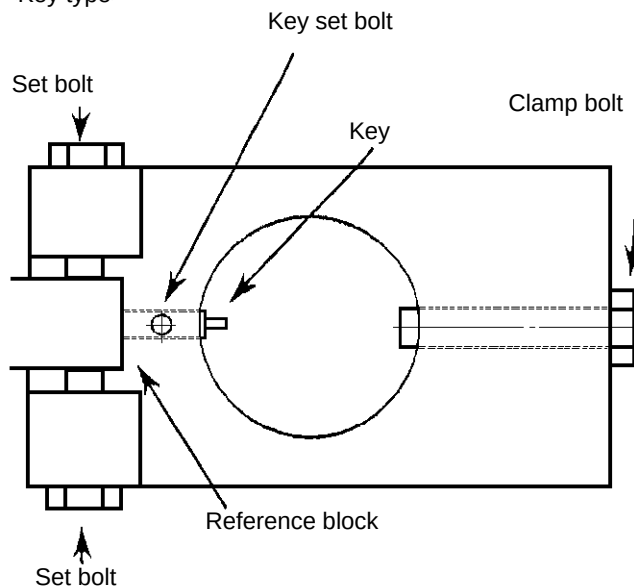
## (b) Changing and Adjusting the hand-operated driving dog

- (1) Set a workpiece in the hand-operated driving dog and tighten the clamp bolts.  
If the key-type driving dog is used, set the key matching the workpiece.
- (2) Place the workpiece, with the hand-operated driving dog, between the centers.
- (3) Adjust the hand-operated driving dog so that its upper face is in parallel with the table upper face by turning the set bolts.
- (4) Tighten the lock nut of the set bolts.

<Pin type>



<Key type>



### <NOTE>

- The set bolt must have wide and flat tip.  
If the set bolt with sharp tip is used, the bolt will be worn soon and the reference block will be damaged.
- Use the reference block, which is locally hardened.
- The gap between the set bolt and the reference lock must be as small as possible (0.02 mm or smaller) since the gap will cause profile error.
- Tightening the clamp bolt will expand the space between the set bolts. Maintain the clamp force constant.

## Revision History

[illegible]

# CHANGING THE SETUP OF WORK SPINDLE

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1-1, Asahi-cho, Kariya, Aichi Pref., 448-8652 Japan

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